

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%
Software: Cisco Packet Tracer, Wireshark

Code of Conduct: *C. Santosh kumar*

Santosh kumar (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *[Signature]*

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Problem Identification	20%				
Debugging	25%				
Optimization	20%				
Analysis	20%				
Code Quality	15%				

Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- IP Address: 192.168.10.65
- Subnet Mask: 255.255.255.192 (/26)
- Default Gateway: 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet 192.168.10.0/26.

Task

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

2. Two devices are configured with the following IPv6 addresses:

Solution

given

IP address : 192.168.10.65

Subnet mask : 255.255.255.192 (26)

Default gateway : 192.168.10.1

Find closest subnet.

A 26 sub was:

Block size = 64

Networks

Network	Host Range	Broadcast
192.168.10.0/26	1-62	63
192.168.10.64/26	65-126	127
192.168.10.128/26	129-190	191
192.168.10.192/26	193-254	255

The IP 192.168.10.65 belongs to

192.168.10.64/26

Network address 192.168.10.0

Broadcast Address 192.168.10.255

Valid Host Range 192.168.10.1 - 192.168.10.254

=> verify IP address.

Configure IP 192.168.10.65 - valid [first usable host]

Correct configuration

Number of usable hosts

Formula $2^{(32 - 26)} - 2$

$$2^6 - 2 = 64 - 2 = 62$$

Solution

1) Address Expansion

Device A - compressed: 2001:db8::ff00:42::8/64
Expanded: 2001:db8:0000:0000:0000:ff00:0042:0008

Device B - compressed: 2001:db8::1/64

Expanded: 2001:db8:0000:0001:0000:0000:0000:0000

2) Network prefix comparison

* Device A/64 prefix: 2001:db8:0000:0000::/64

* Device B/64 prefix: 2001:db8:0000:0001::/64

Optimized configuration & Host Capacity.

Device A : 2001:db8:0:01::60/64

Device B : 2001:db:0:1::1/64

Available Host Address.

$2^64 = 18,446,744,073,709,551,616$ interface

IDs within the 164 subnet

Available Host Addresses:

$$\text{Total host bits} = 128 - 64 = 64$$

Subnet Mask	Prefix	Total ips	Usable Host	Broadcast Address.
255.255.255.192	126	64	62	192.168.1.63
255.255.255.224	127	32	30	192.168.1.95
255.255.255.240	128	16	14	192.168.1.111

Subnet 1 (126)

Network Address : 192.168.1.0

Broadcast Address : 192.168.1.63

Usable Host Range : 192.168.1.1 to 192.168.1.62

Capacity : 62 usable hosts.

Subnet 2 (127)

network Address: 192.168.1.64

Broadcast Address: 192.168.1.63

usable Host Range: 192.168.1.1 to 192.168.1.62

capacity: 62 usable hosts.

Subnet 3 (128)

network Address: 192.168.1.96

Broadcast Address: 192.168.1.111

usable Host Range: 192.168.1.97 to 192.168.1.110

capacity: 14 usable hosts.

Complete Expanded IPv6 Notation (128-bit / 8 Hexets)

device A: 2001:0db8:0000:0000:0000:ff00:0042:0327

device B: 2001:0db8:0000:0000:0001:0000:0000:0001

network prefix (164 - first 4 Hexets)

Device A prefix: 2001:0db8:0000:0000::/64 (compressed)

2001:db8::/64

Device B prefix: 2001:0db8:0000:0001::/64

Compressed 2001:db8:0::/64

not cause / optimized IPv6 configuration.

device A: 2001:db6::1/64

device B: 2001:db6::2/64

subnet mask/prefix 64

calculation of Available Host address.

Host Identifier Bits $128 - 64 = 64$ bits.

Total Available Host:

$$2^{64} = 18,446,744,073,709,551,616.$$

solution

Longest prefix Match Analysis.

destination IP: 192.168.2.100

matches router entry: 192.168.2.0/64.

Next Hop Router.

packets are forwarded to next hop IP: 10.0.0.3.

subnet verification:

* for subnet $192.168.2.0/24$ valid host range is $192.168.2.1$ to $192.168.2.254$.

* $192.168.2.100$ is a valid host range is $192.168.2$ this target subnet.

default route fall back.

if a packet arrives for a destination with no specific match in the table, it will use the default route ($0.0.0.0/0$) and forward to next hop $10.0.0.0$